

BIBEK DHAKAL

Dallas, TX

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Professional Summary

A Computer Science student graduating May 2027 with hands-on experience developing, testing, debugging, and deploying full-stack and backend applications using Java, Python, TypeScript, REST APIs, SQL, and AWS. Experienced with Agile/Scrum, Git/GitHub workflows, unit and integration testing, CI/CD, Docker, and code reviews. Familiar with AWS cloud-native and serverless technologies, including Lambda, API Gateway, S3, DynamoDB, and observability. Strong foundation in software engineering, data structures, OOP, and problem solving, seeking an entry-level Software Engineer role.

Projects

AI Chatbot – Assistance Chatbot | *Java, Maven, JSON*

- Developed a Java-based console chatbot that delivers career guidance on resumes, interviews, internships, and job applications through keyword-based conversation handling.
- Designed a modular architecture using Java Collections and JSON-configured responses, enabling scalable and maintainable conversation management.
- Built and managed the project using Maven while applying object-oriented programming principles and clean software design practices.

ResearchIQ – AI Research Management Platform | *Java, Agile, Git, Senior Design*

- Collaborated in a 5-member Agile team to develop an AI-powered research management platform, implementing and validating core features including authentication, project management, and user dashboards.
- Designed and executed functional and integration tests, identified defects, and verified feature implementations to improve software quality and reliability.
- Participated in sprint planning, backlog refinement, code reviews, and sprint demonstrations while producing technical documentation and sprint deliverables.

MediTrack – Healthcare Appointment & Billing System | *Python, PyTest, SQLite*

- Developed 100+ automated test cases using PyTest to validate patient registration, appointment scheduling, billing, and insurance workflows.
- Applied black-box, white-box, integration, system, and security testing techniques, improving software reliability through comprehensive test coverage and defect detection.
- Created Requirements Traceability Matrices (RTMs), pairwise test suites, and Coverage.py reports to verify requirements compliance and code quality.

Experience

Amazon – Process Guide | *Warehouse Systems, Data Analysis, Process Optimization*

- Utilized Amazon warehouse management and operational software tools to monitor workflow, track inventory movement, analyze productivity metrics, and support daily fulfillment operations.
- Monitored real-time operational data and system dashboards to identify workflow bottlenecks, troubleshoot process issues, and coordinate corrective actions with associates and leadership.
- Supported process optimization by analyzing performance metrics, maintaining accurate system records, and using data-driven insights to improve productivity, quality, and operational efficiency.

Technical Skills

Languages: Java, C, Python, JavaScript, ASP.NET, J2EE, TypeScript, C++, SQL, HTML/CSS

Development: OOP, React, Next.js, Node.js, Express.js, Spring Boot, REST APIs, Bootstrap, Maven, JSON, XML, Android SDK, Google Maps API, Web Application Development

Security and Testing: Secure Programming, JWT Authentication, OAuth, CORS, PyTest, JUnit, Mockito, Postman, Coverage.py, Black-Box, White-Box, Functional, Integration, System, Security, BVA, Equivalence Partitioning, Decision Tables, State Transition, Pairwise Testing, RTM, Unit Testing

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Supabase, Prisma

Software Engineering: Data Structures & Algorithms, Agile Scrum, SDLC, Requirements Analysis, Software Architecture, Design Patterns, CI/CD, Git Version Control, Git Branching, Pull Requests, Deployment Workflows

Cloud & DevOps: AWS, Docker, GitHub Actions, Vercel, Render, AWS Lambda, Amazon API Gateway, Amazon S3, Amazon DynamoDB, Serverless Architecture, Cloud-Native Applications, Amazon ECS, Infrastructure as Code (IaC), AWS CDK, AWS CloudFormation, Terraform, Amazon CloudWatch, Observability, Production Support

AI/ML: Machine Learning, Reinforcement Learning, SVM, Logistic Regression, Naive Bayes, Neural Networks

Tools: Git, GitHub, IntelliJ IDEA, VS Code, Android Studio, Jupyter Notebook, Google Colab, Postman, Linux/Unix

Education

The University of Texas at Arlington

Bachelor of Science in Computer Science

- GPA: 3.76

May 2027

Arlington, TX